

LAB: 07

Topic: Internal and External Commands

List of Commands to Implement:

Date:

Time:

Ver:

Winver:

Cls:

Ipconfig:

Netsh wlan show profile name='____' keyclear:

Logoff:

Start:

Shutdown:

Echo:

Help:

Move:

Copy:

Tree:

Rd:

Ren:

Md:

Cd:

Exit:

Del:

With Description:

Descriptions of the CMD commands you listed:

1. Date

- **Description:** Displays or sets the system date. If used without parameters, it shows the current date and prompts you to enter a new one.

2. Time

- **Description:** Displays or sets the system time. If used without parameters, it shows the current time and prompts you to enter a new one.

3. Ver

- **Description:** Displays the current version of the Windows operating system you're using.

4. Winver

- **Description:** Displays the version of Windows you are running, along with other system information like the build number.

5. Cls

- **Description:** Clears the Command Prompt screen, removing all previously displayed text.

6. Ipconfig

- **Description:** Displays the current network configuration, including the IP address, subnet mask, and default gateway of all network adapters on the machine.

7. Netsh wlan show profile name='____' key=clear

- **Description:** Shows the Wi-Fi profile of a specific network (replace '____' with the actual Wi-Fi network name). The key=clear argument reveals the saved Wi-Fi password if available.

8. Logoff

- **Description:** Logs off the current user session, effectively ending the user's session without shutting down the computer.

9. Start

- **Description:** Launches a program, opens a file, or starts a new instance of a command prompt or other application. For example, start notepad opens Notepad.

10. Shutdown

- **Description:** Shuts down, restarts, or logs off the computer. You can specify various options, like shutdown /s /f to force a shutdown.

11. Echo

- **Description:** Displays a line of text or toggles command echoing on/off. For example, echo Hello displays "Hello" on the screen.

12. Help

- **Description:** Displays a list of available commands in the Command Prompt or provides detailed help for a specific command when used with a command name (e.g., help copy).

13. Move

- **Description:** Moves files or directories from one location to another. For example, move file.txt D:\backup\ moves file.txt to the D:\backup\ folder.

14. Copy

- **Description:** Copies files or directories from one location to another. For example, copy file.txt D:\backup\ copies file.txt to the D:\backup\ folder.

15. Tree

- **Description:** Displays a graphical directory tree of the current folder or a specified path, showing the structure of subdirectories.

16. Rd

- **Description:** Removes a directory. The directory must be empty before it can be removed. For example, rd foldername deletes foldername.

17. Ren

- **Description:** Renames a file or directory. For example, ren oldname.txt newname.txt renames a file from oldname.txt to newname.txt.

18. Md

- **Description:** Creates a new directory (folder). For example, md newfolder creates a folder named newfolder.

19. Cd

- **Description:** Changes the current directory (folder). For example, cd C:\Users changes the current directory to C:\Users.

20. Exit

- **Description:** Exits the Command Prompt or closes the current batch script.

21. Del

- **Description:** Deletes one or more files. For example, del file.txt deletes the file file.txt. Be cautious, as the deleted files cannot be easily recovered.

With Syntax:

1. Date

- **Syntax:**
 - date
 - Displays the current system date.
 - To set the date, just enter the new date when prompted, in MM-DD-YYYY format.
-

2. Time

- **Syntax:**
 - time
 - Displays the current system time.
 - To set the time, just enter the new time when prompted, in HH:MM:SS format.
-

3. Ver

- **Syntax:**
- ver
 - Displays the current version of the Windows operating system.

4. Winver

- **Syntax:**
- winver
 - Displays the detailed Windows version, including the build number.

5. Cls

- **Syntax:**
- cls
 - Clears the screen in Command Prompt.

6. Ipconfig

- **Syntax:**
- ipconfig
 - Displays the IP configuration of the system.
 - **For detailed information:**
 - ipconfig /all
 - Lists all network interfaces and their details (including MAC address, DNS, etc.).

7. Netsh wlan show profile name="____" key=clear

- **Syntax:**
- netsh wlan show profile name="SSID" key=clear
 - Replace "SSID" with the actual Wi-Fi profile name (without the quotes).
 - Reveals saved Wi-Fi credentials (including the password) for the specified network.

8. Logoff

- **Syntax:**
 - logoff
 - Logs off the current user from the system.
-

9. Start

- **Syntax:**
 - start [program_name or file_path]
 - Example:
 - start notepad
 - Starts the specified program or opens the specified file.
-

10. Shutdown

- **Syntax:**
 - shutdown /s /f /t [seconds]
 - /s: Shutdown the system.
 - /f: Forces running applications to close.
 - /t **[seconds]**: Set a timer for shutdown in seconds (e.g., /t 60 to shutdown after 60 seconds).
 - For example, to shutdown immediately:
 - shutdown /s /f /t 0
-

11. Echo

- **Syntax:**
 - echo [message]
 - Displays a message.
 - Example:
 - echo Hello, World!
 - To toggle command echoing on or off:
 - echo on
 - echo off
-

12. Help

- **Syntax:**
 - help
 - Displays a list of available commands in the Command Prompt.
 - **For a specific command:**
 - help [command_name]
 - Example:
 - help copy
-

13. Move

- **Syntax:**
 - move [source] [destination]
 - Moves a file or directory from one location to another.
 - Example:
 - move file.txt D:\backup\
-

14. Copy

- **Syntax:**
 - copy [source] [destination]
 - Copies a file from one location to another.
 - Example:
 - copy file.txt D:\backup\
-

15. Tree

- **Syntax:**
 - tree [drive:\path]
 - Displays the directory tree of the specified path. If no path is provided, it shows the structure of the current directory.
 - Example:
 - tree C:\
-

16. Rd

- **Syntax:**
 - rd [directory_name]
 - Removes an empty directory.
 - Example:
 - rd foldername
-

17. Ren

- **Syntax:**
 - ren [old_name] [new_name]
 - Renames a file or directory.
 - Example:
 - ren oldname.txt newname.txt
-

18. Md

- **Syntax:**
 - md [directory_name]
 - Creates a new directory.
 - Example:
 - md newfolder
-

19. Cd

- **Syntax:**
 - cd [path]
 - Changes the current directory.
 - Example:
 - cd C:\Users
-

20. Exit

- **Syntax:**
 - exit
 - Exits the Command Prompt or ends a batch script.
-

21. Del

- **Syntax:**
 - del [filename]
 - Deletes a file.
 - Example:
 - del file.txt
-

Task to Execute:

1. View System Information

Task:

- Use ver and winver to view the version and build number of your operating system.
- Check the system date and time using date and time.

Commands to Use:

- ver
 - winver
 - date
 - time
-

2. Network Configuration

Task:

- Use ipconfig to view your system's network configuration.
- Use netsh wlan show profile name="YourWiFiNetwork" key=clear to display saved Wi-Fi passwords (if applicable).

Commands to Use:

- ipconfig
 - netsh wlan show profile name="YourWiFiNetwork" key=clear
-

3. Create and Manage Files and Folders

Task:

- Create a new directory, then navigate into it.
- Create a text file, copy it to another folder, and rename it.
- Delete the text file and the directory afterward.

Commands to Use:

- md newFolder
 - cd newFolder
 - echo Hello, World! > file.txt
 - copy file.txt D:\backup\
 - ren file.txt newfile.txt
 - del newfile.txt
 - rd .. (to remove the directory)
-

4. Shutdown or Restart System

Task:

- Use the shutdown command to schedule a system shutdown after 5 minutes.
- Cancel the shutdown.

Commands to Use:

- shutdown /s /f /t 300 (shuts down after 300 seconds)
 - shutdown /a (cancels the scheduled shutdown)
-

5. File Backup Using Copy and Move

Task:

- Create a new folder and copy some files into it.
- Move the files to another location and then remove the original folder.

Commands to Use:

- md backup
 - copy *.txt backup\
 - move backup*.txt D:\backup\
 - rd backup
-

6. View Directory Structure

Task:

- Use tree to visualize the directory structure of a drive (e.g., C:).
- Output the directory structure to a text file.

Commands to Use:

- tree C:\ > directory_structure.txt
-

7. Log Off Current User

Task:

- Use the logoff command to log off the current user session from the Command Prompt.

Commands to Use:

- logoff
-

8. Rename a File

Task:

- Rename a file in a directory using the ren command.

Commands to Use:

- ren file.txt renamedfile.txt
-

9. Clear Command Prompt Screen

Task:

- Use the cls command to clear the screen of the Command Prompt.

Commands to Use:

- cls
-

10. Display All Available Drives

Task:

- Use dir on different drives to view available files and directories in each.

Commands to Use:

- dir C:\

- `dir D:\`
-

11. Use Echo to Create a File with Multiple Lines

Task:

- Use echo to create a text file and add multiple lines of text.

Commands to Use:

- `echo Line 1 > file.txt`
 - `echo Line 2 >> file.txt`
 - `echo Line 3 >> file.txt`
-

12. Modify System Time

Task:

- Use the time command to view and modify the system time.

Commands to Use:

- `time` (view time)
 - To set the time (e.g., 3:30 PM):
 - `time 15:30`
-

13. Run a Program in Background

Task:

- Use start to open Notepad in the background without affecting the Command Prompt.

Commands to Use:

- `start notepad`
-